

Weekly Report: Ultra-High-Pressure ODMR — Does detection-window background flip the green/blue decision?

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Question

Our lock-in sensitivity model $\eta \propto \Delta\nu/(C\sqrt{R})$ counts only NV photons; does adding the spin-independent background collected in the same > 650 nm window (ruby R line, N3/A-band, anvil deformation luminescence) shift the green→blue crossover pressure enough to change which laser we should use at 120 GPa?

Hypothesis

If background dilutes contrast and adds shot noise as $\eta = \eta_0\sqrt{1+\rho}$ with $\rho = B/R$, and blue excitation sits in the Cr^{3+} transmission gap (410/555 nm bands) while shorter blue lines excite N3 more strongly, then the crossover pressure should move with the background-to-signal ratio ρ_0 (defined at 532 nm, ambient) but should not flip green ahead of blue at our 120 GPa target.

Evidence / Interpretation

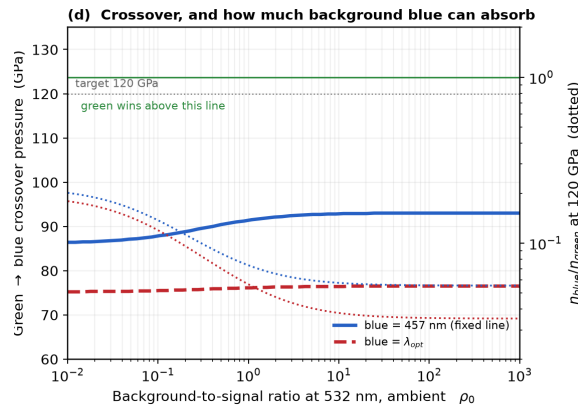


Fig 1: Green→blue crossover pressure vs. background level ρ_0 , for a fixed 457 nm blue line and for the pressure-optimal blue wavelength $\lambda_{opt}(\rho_0)$; right axis: η_{blue}/η_{green} at 120 GPa.

- With 457 nm fixed, the crossover rises from 86.2 GPa ($\rho_0=0$) to 93.0 GPa ($\rho_0 \rightarrow \infty$) because N3 background grows faster toward the near-UV than the Cr^{3+} gap helps at that wavelength.

- Using the wavelength-optimized blue line instead, the crossover stays near 75–77 GPa and η_{blue}/η_{green} at 120 GPa falls from 0.194 to 0.035 as $\rho_0 \rightarrow \infty$, so blue never loses at our operating pressure for any background level.

Gap / Failure

- The channel shapes (ruby U/Y bands, N3 edge, broad luminescence) and their mixing weights are phenomenological placeholders, not fit to our anvil, so the crossover numbers are qualitative rather than calibrated.
- The absolute background level ρ_0 itself is unmeasured, so the model only shows that the conclusion is insensitive to ρ_0 , not what ρ_0 actually is in our cell.

Next Step

Measure the MW-off background spectrum through the > 650 nm long-pass under 405/457/532 nm excitation to pin down ρ_0 and the ruby/N3/broad-luminescence split, which also tells us whether ruby-based pressure calibration should be replaced.

References

1. K. O. Ho, C. Dailledouze, V. Žalandauskas, *et al.*, “Optical Stability and Photophysics of NV Centers in Diamond up to 120 GPa,” arXiv:2606.02399 (2026).
2. A. Dréau *et al.*, Phys. Rev. B **84**, 195204 (2011).
3. Repository: `code/nv_bg.py`, `code/background.py`, `code/fig5_background.py` (this analysis).